Course notes

Differentiating a composite

If u is differentiable on I and v differentiable on an interval containing $u(I)$, then $(v \circ u)' = (v' \circ u)u'$. The domains must be checked before calculating.

$$(u^n)' = nu'u^{n-1}, \quad (e^u)' = u'e^u, \quad (\frac{1}{u})' = -\frac{u'}{u^2}, \quad (\sqrt{u})' = \frac{u'}{2\sqrt{u}} \quad (u > 0).$$

For example, $f(x) = \sqrt{3x+1}$ is differentiable for $x > -\frac{1}{3}$ and $f'(x) = \frac{3}{2\sqrt{3x+1}}$. The sign of f' determines the variations of f; the sign of f do not determine them.

Convexity

A function f is convex on an interval I if its curve lies below its chords: for $x, y \in I$ and $t \in [0,1]$,

$$f((1-t)x + ty) \leq (1-t)f(x) + tf(y).$$

For a twice-differentiable function, we accept the equivalences: f convex; f' increasing; $f'' \geq 0$; the curve lies above all its tangents. The inequalities are reversed for a concave. An inflection point is a point where the curve changes convexity.

Proof: position relative to the tangents. Suppose $f'' \geq 0$. Fix a and let $g(x) = f(x) - f(a) - f'(a)(x-a)$. Since f' is increasing, $g'(x) = f'(x) - f'(a) \leq 0$ for $x \leq a$ and ≥ 0 for $x \geq a$. Thus g reaches its minimum $g(a) = 0$ at a, therefore $f(x) \geq f(a) + f'(a)(x-a)$.

Examples. x^4 is convex and $f''(0) = 0$, but 0 is not the x-coordinate of an inflection point. x^3 changes convexity at 0. The convexity of the exponential function and its tangent at 0 give $e^x \geq 1 + x$.

Continuity and equations

f is continuous at a if $\lim_{x \to a} f(x) = f(a)$. Every differentiable function is continuous. The converse is false: $|x|$ is continuous at 0 but not differentiable at 0.

Intermediate value theorem

If f is continuous on $[a,b]$, any real between $f(a)$ and $f(b)$ has at least one preimage in $[a,b]$. If f is moreover strictly monotonic, this preimage is unique. We can also work on an open interval using the limits at the endpoints.

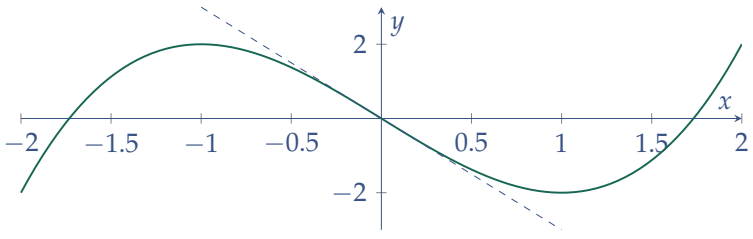
Worked example. $f(x) = x^3 + x - 1$ is continuous on $[0,1]$. Since $f(0) = -1$ and $f(1) = 1$, the intermediate value theorem gives a zero in $]0,1[$. Since $f'(x) = 3x^2 + 1 > 0$, it is unique. $f(0{,}68) < 0 < f(0{,}69)$ provides bounds, without replacing the justifications of existence and uniqueness.

A piecewise-defined function. For $x \neq 1$, set $g(x) = \frac{x^2-1}{x-1}$, and set $g(1) = a$. Since $g(x) = x + 1$ when $x \neq 1$, $\lim_{x \to 1} g(x) = 2$. The function is continuous at 1 if and only if $a = 2$. The point value must therefore be compared with the limit, not replaced by it without justification.

Sequences and continuity. If $u_n \to \ell$ and f is continuous at ℓ, then $f(u_n) \to f(\ell)$. For a recursively defined sequence convergent $u_{n+1} = f(u_n)$, we get $\ell = f(\ell)$.

Bisection

For f continuous and $f(a)f(b) < 0$, calculate $m = \frac{a+b}{2}$. If $f(m) = 0$, stop; otherwise keep the half whose the endpoints have images of opposite signs. After n steps, the length is $\frac{b-}{2^n}$. The midpoint gives an error of at most half the length. The method does not guarantee uniqueness; that is proved separately.



$y = x^3 - 3x$ and its tangent at the inflection point.

Linking calculation, curve and argument

4.1 The inner factor and the domain

First use: S29.

Worked example · A two-step composition

For $f(x) = \sqrt{1+x^2}$, the argument is always strictly positive: f is differentiable on $\mathbb{R}$. The outer function has derivative $t \mapsto 1/(2\sqrt{t})$, the inner function has derivative $2x$. Their product gives $f'(x) = x/\sqrt{1+x^2}$. The fraction is negative for $x < 0$, positive for $x > 0$: f decreases then increases.

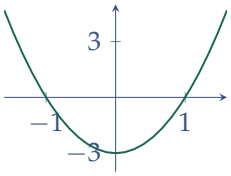
A1 · Immediate application Determine the domain of differentiability and the derivative of $g(x) = \sqrt{2x-1}$ and then $h(x) = e^{1-x^2}$. Check the sign of g'.

A2 · Fix a derivative "For $f(x) = (3x+1)^4$, we have $f'(x) = 4(3x+1)^3$." Explain the factor missing and suggest a check by expanding or using the rate of change at 0.

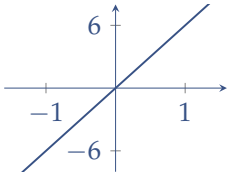
4.2 Associate f, f' and f''

First use: S29 and S31.

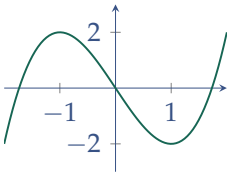
Curve A



Curve B



Curve C



Worked example · A positive decreasing function

For $g(x) = e^{-x}$, g is positive, $g' = -e^{-x} < 0$ and $g'' = e^{-x} > 0$. The curve goes down, but its negative slopes increase towards 0: it is convex. The three pieces of information are consistent.

A3 · Read before calculating The three curves correspond, in an order to be determined, to $f(x) = x^3 - 3x$, f' and f''. Match them using zeros, extrema and changes of convexity; then check by differentiating.

Property · Two readings not to be confused

$f'(a) > 0$ describes a positive slope of f, not a positive y-coordinate. $f''(a) > 0$ indicates locally that the slopes increase; this does not mean that f increases. For example e^{-x} decreases and remains convex.

4.3 Between a tangent and a chord

First use: S31; revisited in S33.

Worked example · Two bounds for x^2

On $[0;2]$, the chord joining $(0;0)$ to $(2;4)$ has equation $y = 2x$; the tangent at 1 has equation $y = 2x - 1$. Convexity gives 2 $x - 1 \leq x^2 \leq 2x$. The differences check out: $x^2 - (2x-1) = (x-1)^2 \geq 0$ and $2x - x^2 = x(2-x) \geq 0$ on this interval.

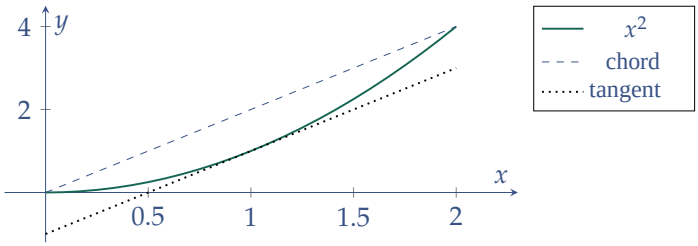


Figure 4.1. Compare the three y-values for $x = \frac{1}{2}$, then justify for every $[0;2]$.

A4 · Compare three functions For x^4, x^3 and $|x|$, study at 0 the continuity, differentiability and change of convexity. For both polynomials, explain why the second derivative vanishing does not is not enough to conclude.

4.4 Existence and uniqueness are two conclusions

First use: S32 and S33.

Worked example · Writing up an approximate solution

Let $F(x) = x^3 + x - 1$. It is a polynomial, so it is continuous. $F(0) < 0 < F(1)$ gives a solution in $]0;1[$ by the IVT. $F'(x) = 3x^2 + 1 > 0$ on $\mathbb{R}$ ensures that there cannot be two. The calculation $F(0{,}68) < 0 < F(0{,}69)$ locates this solution, but does not replace the existence or uniqueness.

A5 · Two counterexamples The function equal to -1 for $x < 0$ and 1 for $x \geq 0$ change sign without vanishing? Which hypothesis of the IVT is missing? For $x^2 = 1$, which assumption is missing for claim a unique solution on $\mathbb{R}$?

4.5 Keep information at every step

First use: S33 and S40.

Property · Bisection invariant

For a continuous function, keep $F(a)F(b) \leq 0$ keeps at least one zero in $[a;b]$. À each step, the length is halved. The algorithm gives a bracketing; uniqueness must be established separately.

Worked example · A first interval

For $F(x) = x^2 - 2$ on [1;2], $F(1) = -1$, $F(2) = 2$. At the midpoint $m = 3/2$, $F(m) = 1/4 > 0$; the half that keeps the change of sign is $[1;3/2]$. Its width is 1/2.

A6 · Choice of strategy Prove that $e^{-x} = x$ has a unique solution in $[0;1]$. Perform two bisection steps, then determine how many steps guarantee an interval width of at plus 1/128.

A first step: Reduce the equation to finding a zero of a function. The number of steps comes from the width, not of the values of the function.

One minute, spoken aloud. State the existence of a solution in one sentence containing the interval, the continuity and the values at the endpoints; state uniqueness in a separate sentence.

Delayed review · S47 then S66. Apply the same arguments to a logarithmic function, then to an area dependent on a bound.

Investigation

Derivatives and reading graphs - 35 min.

Consider the function f defined on $\mathbb{R}$ by $f(x) = x^3 - 3x$. Let $\mathcal{C}_f$ its representative curve.

1. Calculate $f'(x)$ and $f''(x)$, then draw their sign tables.
2. Deduce the variation of f and its intervals of convexity and concavity.
3. Determine the coordinates of the inflection point of $\mathcal{C}_f$ and an equation of the tangent at this point.
4. Using the limits, the extrema and the intermediate value theorem, determine the number of real solutions of $f(x) = 1$.
5. Explain, from this example, why continuity is not enough to ensure the uniqueness of a solution.